16/17 Sem 2 Q3(c)

$$\mathbf{u} \times \mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) \times (\mathbf{w} \times \mathbf{u}) = (\mathbf{u} \cdot \mathbf{v} \times \mathbf{w})^2 .$$

$$\begin{aligned}(\mathbf{v} \times \mathbf{w}) \times (\mathbf{w} \times \mathbf{u}) &= [(\mathbf{v} \times \mathbf{w}) \cdot \mathbf{u}]\,\mathbf{w} \\ &\quad - [(\mathbf{v} \times \mathbf{w}) \cdot \mathbf{w}]\,\mathbf{u} \\ &= (\mathbf{v} \times \mathbf{w} \cdot \mathbf{u})\,\mathbf{w}\end{aligned}$$

$$\begin{aligned}\therefore\ & \mathbf{u} \times \mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) \times (\mathbf{w} \times \mathbf{u}) \\ &= (\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{v} \times \mathbf{w} \cdot \mathbf{u})\,\mathbf{w} \\ &= (\mathbf{v} \times \mathbf{w} \cdot \mathbf{u})\,\underline{(\mathbf{u} \times \mathbf{v} \cdot \mathbf{w})} = (\mathbf{v} \times \mathbf{w} \cdot \mathbf{u})(\mathbf{v} \times \mathbf{w} \cdot \mathbf{u}) \\ & \cancel{= (\mathbf{u} \times \mathbf{v} \cdot \mathbf{w})(\mathbf{u} \times \mathbf{v} \cdot \mathbf{w}) = (\mathbf{u} \times \mathbf{v} \cdot \mathbf{w})^2}\end{aligned}$$

$$\begin{aligned}&= (\mathbf{v} \times \mathbf{w} \cdot \mathbf{u})^2 \\ &= (\mathbf{u} \cdot \mathbf{v} \times \mathbf{w})^2 \quad \#\end{aligned}$$